



A discrete choice model based on random utilities for exit choice in emergency evacuations



Ruggiero Lovreglio^a, Dino Borri^a, Luigi dell'Olio^{b,*}, Angel Ibeas^b

^a Department of Civil and Environmental Engineering, Technical University of Bari, Italy

^b Department of Transport and Logistic, University of Cantabria, Santander, Spain

ARTICLE INFO

Article history:

Received 23 May 2013

Received in revised form 21 August 2013

Accepted 5 October 2013

Available online 25 October 2013

Keywords:

Evacuation

Emergency situations

User behaviour

Mixed logit

Stated preference survey

ABSTRACT

This work presents a behavioural model based on discrete choice models (DCMs) in order to simulate decision maker behaviour during an emergency situation. The proposed model is a probabilistic exit choice model that allows understanding the heterogeneity of the different decision maker's tastes. A stated preference survey was designed and realized with a sample of decision makers in order to achieve this objective. Thus, the obtained data was used to model the optimal mixed logit model. The use of DCMs allows to know the behaviour of different decision maker types during an evacuation process. The case study highlights the importance of the influence of other decision makers on the decision-making process. This work could be used as the starting point in the development of behavioural models which could be implemented in the current tools for the simulation of emergency evacuation.

© 2013 Elsevier Ltd. All rights reserved.

1. Introduction

Over the last few decades, Performance Based Design (PBD) has been introduced into the different codes of several countries in order to give designers a flexible and effective method of evaluating safety levels in structures and infrastructures during emergency situations (Meacham et al., 1998; Hadjisophocleous and Bénichou, 2000; Lo et al., 2004; Tavares, 2009; Ronchi et al., 2013; Lovreglio et al., 2013). In fire engineering terms, PBD consists of the comparison between two different times: required safe egress time (RSET) and available safe egress time (ASET) (Nelson et al., 2002; Purser, 2003).

PBD allows designers to use predictive evacuation models, which are able to reproduce real evacuation processes and therefore, provide insights into human behaviour under different emergency situations (Kuligowski et al., 2010; Bensilum, 2003).

Evacuation behaviour is complex because it is characterized by both physical factors, such as the user's movements towards a safe space, and psycho-social factors (Santos and Aguirre, 2004). Ronchi et al. (2012a,b) splits these factors into internal and external, the relevant literature shows that the most important external factors are social interaction (Nilsson, 2009; Kuligowski et al., 2011; Purser and Bensilum, 2001; Frantzych et al., 2001; Sime, 1985) and the prevailing environmental conditions (such as warning systems, and visibility) (Nilsson, 2009; Norén and Winér, 2003).

The main social interaction, which has a fundamental role during evacuation, is the *social influence* (Latané and Darley, 1968; Nilsson and Johansson, 2009; Nilsson, 2009). According to the literature, social influence can be divided into normative and informational influence. The first influence leads people to behave in accordance with their expectations of other individuals. The second influence occurs when people's understanding of a situation is influenced by the action or inaction of others (Deutsch and Gerard, 1955; Nilsson and Johansson, 2009; Kinatader, 2012).

A number of internal factors can be used to characterize human behaviour, such as: physical abilities (which may depend on age or health) and socio-psychological characteristics (direct or indirect risk perception, emotional states, cultural background or training, past experiences, familiarity with the environment and *affiliation* behaviour (Ronchi et al., 2012a,b; Fridolf et al., 2011; Galea et al., 2010; Gandit et al., 2008; Worm et al., 2006; Wilde, 2001; Sime, 1985). In particular, *affiliation* behaviour occurs when people are attracted to and move towards familiar persons and places (Sime, 1985).

The complexity of human behaviour during emergencies led to the development of a variety of evacuation models (Kuligowski et al., 2010; Johnson, 2005; Gwynne et al., 1999). Those models usually calculate the aforementioned RSET in accordance with the simplified time line model (BS PD7974, 2004; Purser et al., 2007; Candy et al., 2006).

One of the most important issues of these models is route choice modelling and exit selection (Ronchi et al., 2012a,b).

The literature agrees that the exit choice is not only influenced by the perceived time required to evacuate from a specific exit

* Corresponding author. Tel.: +34 942202262; fax: +34 942201703.

E-mail address: dellololi@unican.es (L. dell'Olio).

from those available. In fact, familiarity with the exit according to *affiliation theory* (Sime, 1985; Proulx, 1993; Pan, 2006), herding behaviour (Helbing et al., 2002; Pan, 2006; Low, 2000) and the effect of cooperative or selfish behaviour according to *social influence* (Cirillo and Muntean, 2012; Heliövaara et al., 2012; McLean et al., 1996; Muir et al., 1995), and environmental conditions (Nilsson, 2009; Ronchi et al., 2012a,b) can also influence the choice process. A multiplicity of models are available in the literature for the study of this process (Kuligowski et al., 2012). In some models the agent selects the nearest exit without any consideration of other factors, while in other models the agent selects the exit on the basis of optimal decision (shortest amount of time) or consideration of an environmental condition (other occupants' behaviours, fire, etc.) (Kuligowski et al., 2012; Heliövaara et al. 2012).

Random Utility Models (RUMs) make an important contribution to the modelling process and allow predicting the behaviour of individuals in choice situations (Ben-Akiva and Lerman, 1985). In fact they allow the uncertainty of human behaviour to be embedded in one or more random components (McFadden and Train, 2000). Another advantage of these models is that choices are also influenced by the decision maker's characteristics (Greene, 2000). To cite specific applications, RUM modelling of pedestrians walking has addressed the 'short range' (walking) behaviour of individuals (Antonini et al., 2005; Robin et al., 2011): these works are based on a decision making framework initially proposed by Hoogendoorn et al. (2002) and Daamen et al. (2004), which shows the hierarchical structuring of choices. According to this view, decision making can be split by cascade into a *strategic* level (the users choose the strategy to be performed), a *tactical* level (the users choose the tactical actions to be performed on the basis of the strategic level and schedule their actions accordingly), and an *operational* level (the users take snap decisions in order to instantiate the choices made in the two previous levels) (Hoogendoorn et al., 2002; Daamen et al., 2004; Hoogendoorn and Bovy, 2004). In an evacuation context the first and highest level coincides with the decision to go towards a safe place, which is the fundamental choice made during the *pre-movement time* (Purser et al., 2007), the second level coincides with the process of route and exit selection and affects the *movement time* (Candy et al., 2006), in the end, the third level embeds the typical short range choices which characterize the evacuation process and also affects the *movement time* (Antonini et al., 2005; Robin et al., 2011).

The present work focuses on the tactical level and in particular on the modelling of the exit choice process during emergency evacuation situations. We investigate the exit choice processes which are seen as fundamental elements in emergency evacuation routing.

Among all the factors listed above, this work focuses on how the following defined factors influence exit choice: the decision maker's proximity to an exit, the crowds near exits, herding behaviour and the effect of cooperative or selfish behaviour.

The main contributions made by this work are: firstly, the development of an exit choice model for emergency evacuation behaviour based on the RUMs approach; secondly, the consideration of human factors in order to achieve more detailed modelling of human behaviour based on the inclusion of the systematic and random tastes of different decision makers.

The first part of the paper analyses the evolution of RUMs and stated preferences survey techniques, in the light of the advantages they offer compared with the current simplified mainstream models. The second part describes a case study. The third part involves a discussion of the pros and cons coming from the results. The paper ends with some suggestions about future work that is needed to overcome the persistent limitations of our approach in challenging the complexity of emergency evacuation behaviour (EEBs).

2. State of the art

RUMs belong to the DCM (Discrete Choice Models) class of models which are used to model choices taken by decision makers when they are put in front of a finite set of alternatives (Train, 1986).

This theory was introduced in the scientific literature by Thurstone (1927), who considered the usefulness perceived by the "q" decision maker about the "i" alternative represented by the following sum:

$$U_{iq} = V_{iq} + \varepsilon_{iq} \quad (1)$$

This equation highlights that the utility function comes from the sum of two terms. The first one (V_{iq}) is a systematic quantity, meaning the main or expected value of the perceived utility. The second term (ε_{iq}), called the *random residual*, represents the deviation of the average utility from the real value. In this way all the factors that make the decision-making model deviate from pure rationality are embedded in the random residual. The scientific literature provides various models of random utility which differ from each other because of the different probability distributions chosen by different authors for modelling the random residual (Cascetta, 1998).

The Mixed Logit Models (MLMs) used in this paper (Boyd and Mellman, 1980; Cardell et al., 1980) are based on the hypothesis that the random residues (ε_{iq}) are independent and identically distributed (IID) according to a Gumbel random distribution with a mean equal to zero and a λ parameter (McFadden, 1974). MLMs are highly flexible and can approximate any random utility model (McFadden and Train, 2000) overcoming the limitations of standard Logit models (Train, 2009). In fact, MLMs can model the decision maker's varying tastes by the use of random distributions for the θ_{ik} coefficients and the probability of choosing a certain alternative can be provided by the following integral:

$$MLP_{jq} = \int P_{jq}(\theta) f(\theta | \beta) d\theta \quad (2)$$

where P_{jq} has the expression of Multinomial Logit probability, while θ is the vector of the generic values that are assumed by the θ_{ik} coefficients and have $f(\theta | \beta)$ probability. Finally, β is the vector of the parameters characterizing the probability distribution f . Unlike Multinomial Logit and Nested Logit Models, MLM do not have closed solutions (see Eq. (2)). Nevertheless, solutions can be achieved both by numerical integration and by using Monte Carlo draws ϑi ($i = 1 - R$) from $f(\vartheta | \beta)$ based on the Halton sequence and the following equation (Hensher, 2001; Hensher et al., 2005):

$$MLP_{jq} = \frac{1}{R} \sum_{i=1}^R P_{jq}(\vartheta i) \quad (3)$$

To date, RUMs have been widely used in econometrics and have had many useful applications in a diverse range of fields: marketing, finance, etc. These models have also been used by different authors to solve transportation issues (Ben-Akiva and Lerman, 1985; Cascetta et al., 1992; Antonini et al., 2005). Their fields of application have been various, for example, they have been used to model transport demand (dell'Olio et al., 2009) and to evaluate the public's perception of quality in a public transport service (dell'Olio et al., 2011).

This paper demonstrates the utility of RUMs in modelling the process of choosing an emergency exit during emergency conditions. In the literature, the modelling of exit choice has been addressed in different ways (Kuligowski et al., 2012; Heliövaara et al. 2012). For example Lo et al. (2006) and Ehtamo et al. (2010) use game theory, while other authors such as Heliövaara (2007) used deterministic models based on utility maximization by decision makers. A deterministic approach to the problem has

the limitation of only being able to simulate average decision-maker behaviour, rather than taking advantage of the differences and uncertainties inherent in the choice process. Alternatively, RUMs can embed these uncertainties in order to provide results that best suit the real processes of choice. This feature is addressed and discussed here to add value to current research as well as to improve the models that are currently being used. Moreover, RUMs show the advantage of providing formulations that depend not only on the variables characterizing the environment but also on the characteristics of the decision-makers themselves and the variability of their tastes. On the other hand, deterministic models have a closed formulation which is easily implemented in software whereas the more complex RUMs (i.e. MLMs) can present integral expressions requiring numerical solutions that increase the computational burden (see Eq. (3)).

3. Methodology

RUMs can be approached in different ways. The approach used in this paper is inspired by the one proposed by [dell'Olio et al. \(2011\)](#). All the steps performed can be brought together in three main areas: Background, Survey Design and Data Collection and Modelling ([Fig. 1](#)).

The first step is to define the problem in order to locate all the factors and variables which characterize it. This step requires a preliminary survey of the existing literature, followed by Focus Group (FG) techniques which help to avoid missing any implicated variables or factors. FGs are generally complex and may require their own methodological framework such as the one proposed by [Ibeas et al. \(2011\)](#).

The second step involves the development of the survey. The survey is divided into two parts. The first part involves a demographic survey to collect all the information characterizing the interviewee. This information is useful during the modelling phase; in fact the utility functions could also depend on the intrinsic characteristics of the decision maker. The second stage

involves the application of different scenarios which were designed based on the variables relating to the issue. In this work *fractional factorial design* is used to perform the second part of the survey and an orthogonal scenario design is made following the tabular method proposed by [Kocur et al. \(1981\)](#).

The last step involves the determination of the optimal model based on the answers given by the interviewees using statistical criteria coming from the scientific literature (Reference).

4. The case study

The present work starts with a bibliographic review of exit choice during emergency situations. There are a multiplicity of factors influencing the decision making process which have already been discussed in the introduction section ([Kuligowski et al., 2012](#); [Heliövaara et al., 2012](#); [Ronchi, 2012](#); [Nilsson, 2009](#); [Fridolf, 2010](#); [Latané and Darley, 1968](#); [Sime, 1985](#); [Proulx, 1993](#); [Pan, 2006](#); [Nelson et al., 2002](#); [Pan, 2006](#); [Low, 2000](#); [McLean et al., 1996](#); [Muir et al., 1995](#)). The preliminary framework was taken from the literature and later complemented by a focus group held

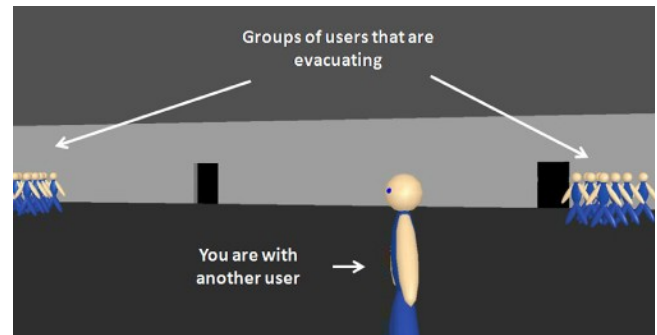


Fig. 2. Frame from one of the administered videos.

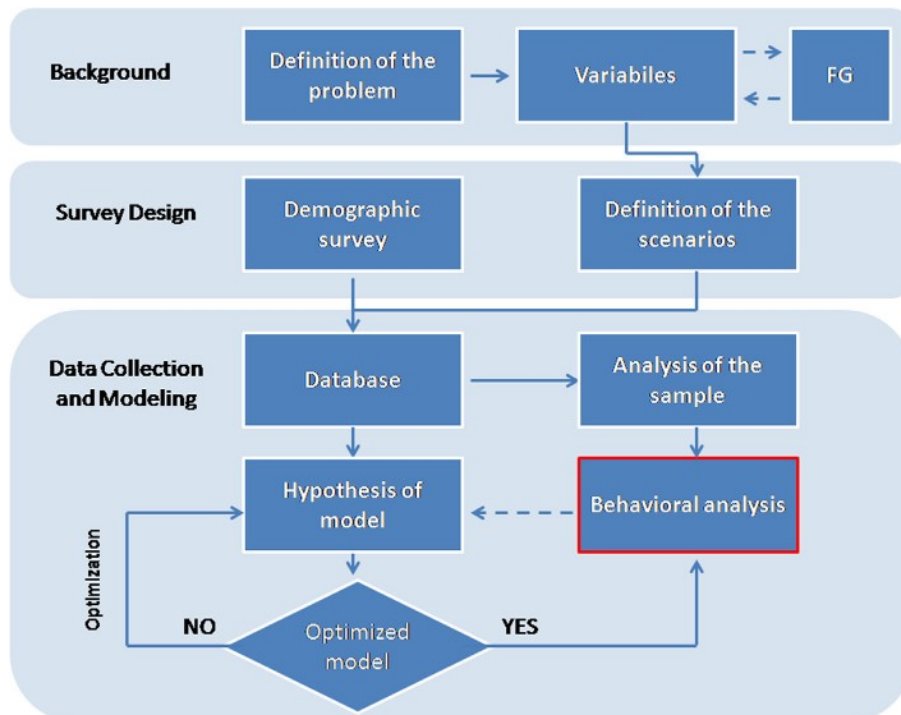


Fig. 1. Methodological scheme (inspired by [dell'Olio et al. \(2011\)](#)).

Table 1
Levels for each variable.

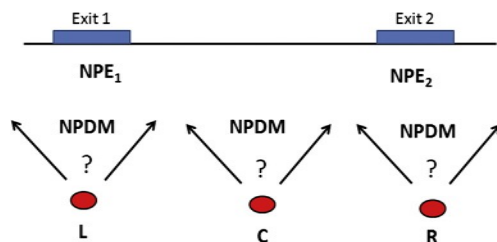
Level	NPE ₁	NPE ₂	NPDM	NEAREX ₁	NEAREX ₂
1	0	0	0	0	0
2	5	5	1	1	1
3	10	10	4	/	/
4	20	20	8	/	/

Table 2
Orthogonal design.

Scenario	NPE ₁	NPE ₂	NPDM	NEAREX ₁	NEAREX ₂
1	0	0	0	0	0
2	0	5	1	1	0
3	0	10	4	1	1
4	0	20	8	0	1
5	5	0	1	1	1
6	5	5	0	0	1
7	5	10	8	0	0
8	5	20	4	1	0
9	10	0	4	0	1
10	10	5	8	1	1
11	10	10	0	1	0
12	10	20	1	0	0
13	20	0	8	1	0
14	20	5	4	0	0
15	20	10	1	0	1
16	20	20	0	1	1

Table 3
Decision maker's possible position.

NEAREX ₁	NEAREX ₂	Positions
1	1	Center (C)
1	0	Left (L)
0	1	Right (R)
0	0	Center (C)

**Fig. 3.** Decision maker's possible positions.**Table 4**
Final orthogonal design.

Scenario	NPE ₁	NPE ₂	NPDM	Position
1	0	5	1	L
2	20	10	1	R
3	0	10	4	C
4	10	10	0	L
5	20	5	4	C
6	0	20	8	R
7	10	20	1	C
8	5	5	0	R
9	10	5	8	C
10	5	20	4	L
11	5	0	1	C
12	10	0	4	R

at an Italian university (Politecnico di Bari, Italy) which did not provide any other factors.

At this stage the case study has been simplified by only considering a few of these variables. The main goal of this work is to understand how the *social influence* (in particular, herding behaviour, the effect of cooperative or selfish behaviour) and proximity to an emergency exit affect the choice being made while the *affiliation* behaviour can not be investigated because of the impossibility of including familiar people or places in the studied virtual environment.

In the case study the interviewees can choose between two alternatives corresponding to two emergency exits with similar characteristics. Thus if “i” is the variable representing the exit, then it can take on the values 1 or 2. The other variables embedded in this work are:

- The number of decision makers next to an exit (useful in studying the influence of the choices made by crowds near exits and the herding behaviour).
- The number of decision makers next to the interviewee (useful in studying how cooperative or selfish behaviour influence choice).
- Proximity to a safe exit.

4.1. The survey

The survey was based on a questionnaire delivered through the internet over a period of three weeks and was presented to students from different degree courses at the Politecnico di Bari, Italy. Furthermore, the questionnaire was also publicized outside the university through the use of e-mail and social networks.

The survey is divided into two main parts.

The first part of the survey contains both descriptions about the survey and a preliminary demographic questionnaire and is divided into three web pages:

1. The first page shows a brief description in order to provide the necessary information to fill it in. This part deliberately does not show a great deal of detail to avoid influencing the interviewees in any way.

Table 5
Optimal model.

Variable	Coefficient	Test-t
<i>Random parameter*</i>		
NPE	−0.148	−13.790
NEAREX*HIGH	0.007	3.130
ASC	0.318	2.836
<i>Non random parameter</i>		
NPDM	−0.130	−3.531
NPDM*GR	0.081	2.381
NEAREX*AGE	−0.024	−1.668
<i>Diagonal values in Cholesky matrix L</i>		
NPE	0.097	8.263
NEAREX*HIGH	0.005	3.075
ASC	0.020	0.022
<i>Below diagonal values in L matrix $V = L \cdot L^T$</i>		
NEAREX*HIGH-NPE	0.003	2.162
ASC-NPE	0.033	0.235
ASC-NEAREX*HIGH	0.462	2.330
<i>Standard deviation of parameter distribution</i>		
Sd NPE	0.097	8.263
Sd NEAREX*HIGH	0.006	4.007
Sd ASC	0.464	2.339
$L(\theta)$	−1195.143	
Normal distribution for NPP, VICIN*HIGH and ASC		

Table 6

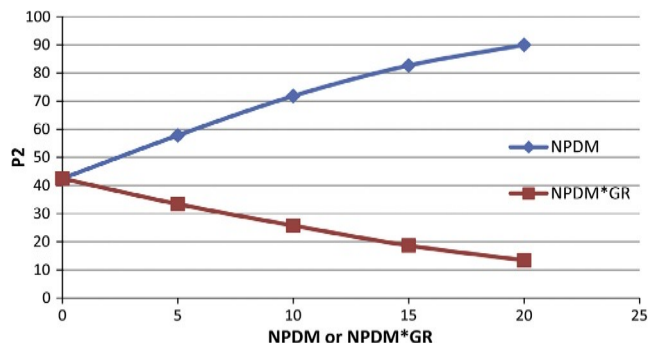
Influence of NPDM on choice.

P_1 (%)	P_2 (%)	NPE ₁	NPE ₂	NPDM	NPDM*GR	NEAREX*HIGH	NEAREX*AGE
57.51	42.49	10	10	0	0	0	0
42.18	57.82	10	10	5	0	0	0
28.18	71.82	10	10	10	0	0	0
17.34	82.66	10	10	15	0	0	0
10.02	89.98	10	10	20	0	0	0

Table 7

Influence of NPDM*GR on choice.

P_1 (%)	P_2 (%)	NPE ₁	NPE ₂	NPDM	NPDM*GR	NEAREX*HIGH	NEAREX*AGE
57.51	42.49	10	10	0	0	0	0
66.59	33.41	10	10	0	5	0	0
74.25	25.75	10	10	0	10	0	0
81.32	18.68	10	10	0	15	0	0
86.59	13.41	10	10	0	20	0	0

**Fig. 4.** Influence of NPDM and NPDM*GR on the choice of the right hand exit.

2. The second page involves the administration of the demographic questionnaire by which a characterization of the interviewee can be made. The demographic survey contains 11 questions about the interviewee's physical and working characteristics (sex, age, education, job, family income, weight, height, relevant handicap, emotional state, work related to evacuation issue) and about their own experiences with evacuations during emergency situations. In particular, the interviewee must provide information about different aspects of their own rationality:

- Emotional assessment (calm/anxious).
- Analytical assessment (impulsive/reflective).
- Social assessment (introvert/extrovert).

All this information is useful during the modelling phase because it refers to variables of interaction which are required to characterize the utility functions of the decision-makers according to their socio-economic characteristics.

Table 8

Influence of NEAREX*HIGH on choice.

P_1 (%)	P_2 (%)	NPE ₁	NPE ₂	NPDM	NPDM*GR	NEAREX*HIGH	NEAREX*AGE
51.58	48.42	10	10	0	0	150	33
49.93	50.07	10	10	0	0	160	33
48.32	51.68	10	10	0	0	170	33
46.75	53.25	10	10	0	0	180	33
45.23	54.77	10	10	0	0	190	33
43.76	56.24	10	10	0	0	200	33

1. The third page continues with an additional page of instructions to introduce the interviewee to the main body of the questionnaire covering the scenarios about which they will be asked to choose. The interviewee is informed that some videos will be shown and that they must make a choice at the end of these videos. Some instructions on how the interviewee should address the choice process are also provided. In this regard, the following statement is essential:

*“Bear in mind that **there is no correct answer to the questions** being asked, so please respond according to how you would choose during a real emergency.”*

This statement represents a key part of the survey. In fact, the goal of this work is not to identify what could be the most rational behaviour in the choice process but the real behaviour of interviewees during an emergency situation.

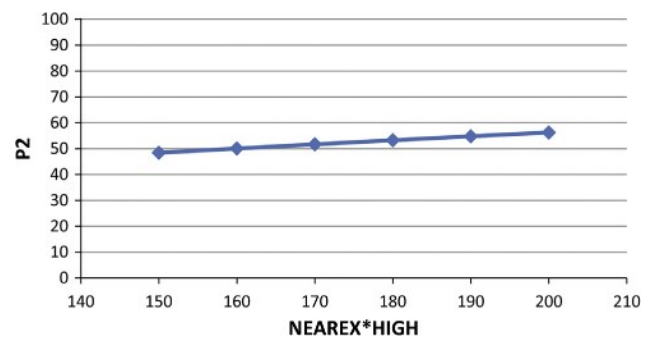
**Fig. 5.** Influence of NEAREX*HIGH on the choice of right hand exit.

Table 9
Influence of NEAREX*AGE on the choice.

P_1 (%)	P_2 (%)	NPE ₁	NPE ₂	NPDM	NPDM*GR	NEAREX*HIGH	NEAREX*AGE
41.2	58.8	10	10	0	0	170	20
46.66	53.34	10	10	0	0	170	30
52.21	47.79	10	10	0	0	170	40
57.7	42.3	10	10	0	0	170	50
63.02	36.98	10	10	0	0	170	60
68.05	31.95	10	10	0	0	170	70

The second part of the survey involves showing the interviewee 12 videos and asking them to make a choice at the end of each one (see Video 1). These videos are recorded thanks to FDS + Evac (Korhonen et al., 2010) and always show two emergency exits in a closed environment but with some variations:

- The position of the interviewee in the closed environment.
- The number of people present in the vicinity of the two exits.
- The number of people present in the vicinity of the interviewee.

Fig. 2 shows a frame from one of these videos.

The following variables (or attributes) are used in order to define the scenarios shown by these video:

- NPE_i number of people in the vicinity of “i” exit.
- NPDM number of people in the vicinity of decision maker.
- NEAREX_i Boolean variable, that is equal to 1 if the observer is placed near the “i” exit and 0 if otherwise.

The first three variables have 4 levels while the last two have only two levels per definition. Table 1 shows the levels for each variable.

The total number of possible scenarios is $3^4 2^2 = 324$. Thus, the orthogonal scenarios are chosen according to the tabular model by Kocur et al. (1982) (Table 2).

The combinations of the values of NEAREX₁ and NEAREX₂ characterize the position of the decision maker. In fact, there are four possible combinations to which the positions shown in Table 3 correspond.

Fig. 3 also shows the three positions that the decision maker can take.

The first and last scenarios from Table 2 are eliminated because there are two identical alternatives. Moreover, scenario 4 is symmetrical to 13 while scenario 7 is symmetrical to 10, so scenarios 7 and 13 are deleted. Thus, the remaining 12 scenarios are arranged in order to ensure a succession of more dissimilar scenarios. The succession used for the survey is shown with updated numeration in Table 4.

The first video is shown to the interviewee at the start of the final part of the survey. At the end of the video the interviewee is automatically redirected to a new web page where the interviewee has to choose between the left and right exits. This page shows the interviewee a countdown timer that gives them only 5 s to answer the question. The counter is a useful device to avoid too much rationality in the decision making process. Actually the web page has been programmed so that the user is still able to select the exit after the time has elapsed to avoid the database being loaded with scenarios devoid of choice. However, the interviewee is unaware of this aspect in order to avoid their alertness threshold being lowered.

After making their choice, the interviewee is directed to the second video and the process is repeated for each video.

At the end of the sequence of videos and related questions, the user is automatically directed to a web page providing confirmation of the registration of their data and thanking them for taking part in the survey.

4.2. Results and limitations

The survey collects 191 observations. From a brief analysis, a sample seems not sufficiently heterogeneous for some variables. The main information about this homogeneity is outlined below:

- 31% female and 69% male.
- Average age is equal to 25 and the standard deviation is equal to 4.4.
- The majority of the sample has a high-medium level of education (63% of the sample were graduates and 35% of the sample stated they had finished high school) and declared they were students (83%).

Even if the data collected constitutes a valuable knowledge base for modelling, some of the above-mentioned homogeneities could influence the estimated model.

Another limitation of this methodology is that the stated choice may not coincide exactly with those found in a real emergency situation. This is due to the impossibility of exactly recreating a real emergency situation with its relevant physiological implications (i.e. stress) through an online survey.

Moreover, the videos used show a simplification of a real choice situation during an emergency through a virtual environment that could reduce the social influence according to the threshold model of social influence (Blascovich, 2002). In fact, a decision maker could not truly perceive the simulated agents as real persons because of the low behavioural realism and agency (Kinatader, 2012; Blascovich, 2002). Future research may investigate the reliability of the results obtained through the methodology in use. This includes the assessment of the degree of perceived realism of the virtual environment (Held and Durlach, 1992; Kinatader, 2012), which is instead not considered in this paper.

The integration of revealed preferences into real or simulated (i.e. large-scale experiments) emergencies could also be useful in improving the knowledge base used in this work.

5. Modelling

A Step-wise Fashion procedure is used to find the optimal model shown in Table 4. This is a mixed Logit model which embeds

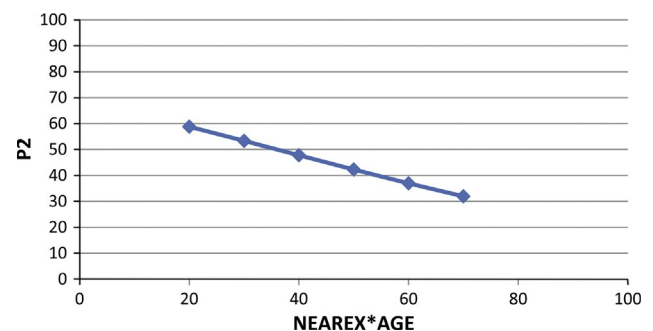


Fig. 6. Influence of NEAREX*AGE on the choice of the right hand exit.

three random parameters with normal distributions (NPE, NEAREX*HIGH and ASC). The model also embeds systematic variations in decision-maker tastes (interactions of the proximity to the exit variable with height and age and the number of people near the exit to the standard of education) and correlations between random parameters. It can also be considered as a model with panel data because it considers the correlation between the replies of individuals who responded to the different scenarios proposed in the stated preference investigation (Traun, 2009). A number of 200 Halton draws are used to simulate the probability of choice (random distribution of the parameters) (Train, 1999; Hensher, 2001; Hensher et al., 2005).

A more detailed interpretation of the model highlights that the choice of the exit during emergency situations is influenced by service level variables (the number of people near the exit and closeness to the person who is about to evacuate), the proximity of the exit and socio-economic variables that represent the characteristics of the person involved in the evacuation process (such as their age, standard of education and height). The choice does not seem to depend on the emotional state declared by the respondents which could be because the data collection method (online state survey) used is not able to emotionally involve the decision-maker due to the limitations presented in Section 4.2. Another hypothesis could be that emotional state takes second place in the emergency situation. In fact, when the interviewee lives the emergency in the experiment (see Video 1), they are overwhelmed by increasingly important variables, the effect of group rather than individual behaviour (people involved in the evacuation phenomenon are influenced by the number of people around them and the people who are near the exit). This behaviour is important because it helps in understanding that evacuation models should simulate group dynamics rather than only addressing individual behaviour because *social influence* causes this individual behaviour to change in relation to the behaviour of other evacuees.

Another important observation is related to the heterogeneity of decision-maker behaviour. In fact, this model allows modelling the tastes of different decision-makers by using socio-economic variables as they differ from one another through their different socio-economic characteristics.

The model allows the correlation of the random variables defining behavioural heterogeneity that is not explained by the interactions with the socio-economic variables. This provides clearer information about decision maker behaviour (Table 5, “Below diagonal values in L matrix. $V = L^*LT'$ ”). The case study shows a significant correlation between the proximity to the exit, the height of the person involved in the evacuation and the number of people near the door (NEAREX*HIGH-NPE) and between the specific constant of the alternative, the proximity of the exit, the height the person involved in the evacuation.

The presence of the alternative specific constant (ASC) is important because it picks up all the effects of the variables that the modeller has not included in the model but which might be relevant. This constant is considered as a random variable and this is very important because this allows the analysis of the changing behaviour of the decision makers due to the variables that were not initially considered. The following section describes a sensitivity analysis performed on the model in order to better understand the results.

5.1. Sensitivity analysis and behavioural analysis

A sensitivity analysis quantifies how individual variables affect the probability of choosing an exit. The influence of NPE_1 and NPE_2 has not been studied because it is clearly visible from the model. In fact, the model shows that decision makers generally perceive other users near to exits as a disutility although the probabilistic

approach used in this work allows us to evaluate the probability that a decision maker shows herding behaviour.

The influence of NPDM and NPDM*GR (from the first utility function) is evaluated first. Both these variables cannot take non-zero values, indeed, while NPDM quantified how a decision maker generally perceives the presence of other people close to him, the variable NPDM*GR quantifies the same object in the case in which the decision maker is a graduate. The results are shown in Tables 6 and 7 and in Fig. 4.

Thus, the behaviour of the decision maker according to the number of people around them varies as a function of their standard of education. The scenarios considered in Tables 5 and 6 include the decision maker not being near to the right hand exit and an equal crowding of both exits. In this case a generic decision maker tends to prefer the left hand exit (which is closest to them) when they are alone (NPDM = 0), although they prefer the right hand output as NPDM increases. Thus, in this latter context the decision maker perceives the NPDM variable as a disutility for the left hand exit. This is probably due to the fact that a decision maker considers that nearby generic users tend to discard the right hand exit (that is not close to them) and therefore consider it to be free and more accessible. Thus, the decision maker would seem to depart from the choice of the group next to him by showing a self-ish behaviour however, graduate decision makers reveal an opposing tendency. This might be explained by the greater confidence of the graduate decision maker in the choice of the nearby group in accordance with cooperative behaviour.

Table 8 and Fig. 5 show the analysis of the influence of NEAREX*HIGH on choice.

In the analysed scenarios the decision maker is near the right hand exit and has an age of 33 years while the two exits have almost the same number of users. These scenarios show that decision makers show a growing interest in the nearest exit as their height increases. This trend could be explained by the fact that a user has a higher visibility of what is happening near the door closest to them. However, none of the videos used in the survey provide a different point of view of the scene as a function of the height of the interviewee, so further research is required to understand this trend more fully.

Table 9 and Fig. 6 show how NEAREX*AGE influences the choice.

In the analysed scenarios the decision maker is located near to the right hand exit and has a height of 170 cm, while the two exits have almost the same number of users. In these scenarios the decision makers tend to choose the farthest exit with increasing age. Further research is also required to obtain a fuller understanding of this trend.

6. Conclusions

The main goal of this paper was to define an exit choice model for emergency situations. In this work, a Mixed Logit framework is used for a probabilistic model that deviates from the typical stiffness of the semi-deterministic or deterministic models and integrates intrinsic uncertainty as inherent in the process. The model is then completed by the use of data collected from an online declared preference survey.

The methodology highlighted some of the variables and factors which influence the selection process, how they affect the choice made and the heterogeneity of the decision makers. In fact, both the randomness in decision maker tastes and the systematic variations coming from the socio-economic characteristics of the decision-makers themselves (standard of education, height and age) have also been incorporated into the model. The resulting model has shown the importance of the influence of group dynamics (herding behaviour, cooperative or selfish behaviour) because it

changes as a function of the behaviour of others participating in the evacuation process. Although the model does not neglect the effects of variables that have not been explicitly mentioned by a constant with its own random distribution, further experimental research is required in order to acquire a complete knowledge about exit choice behaviours. A more complex methodological framework (Walker and Ben-Akiva, 2002; Ben-Akiva et al., 2002) can be used to further investigate the psychological and environmental factors influencing this issue.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ssci.2013.10.004>.

References

- Antonini, G., 2005. A discrete choice modeling framework for pedestrian walking behavior with application to human tracking in video sequences, PhD Thesis, École Polytechnique Fédérale de Lausanne, Switzerland.
- Bensilum, Purser, 2003. GridFlow: an object-oriented building evacuation model combining pre-movement and movement behaviours for performance-based design. *Fire Safety Science – Proceedings of the Seventh International Symposium*, 941–953.
- Ben-Akiva, M., McFadden, D., Train, K., Walker, J., Bhat, C., Bierlaire, M., Bolduc, D., Boersch-Supan, A., Brownstone, D., Bunch, D., Daly, A., De Palma, A., Gopinath, D., Karlstrom, K., Munizaga, A., 2002. Hybrid choice models: progress and challenges. *Marketing Letters* 13 (3), 163–175.
- Ben-Akiva, M.E., Lerman, S.R., 1985. *Discrete Choice Analysis: Theory and Application to Travel Demand*. MIT Press, Cambridge.
- Blascovich, J., 2002. A theoretical model of social influence for increasing the utility of collaborative virtual environments. In: *Proceedings of the 4th International Conference on Collaborative Virtual Environments*, Bonn, Germany.
- Boyd, J., Mellman, J., 1980. The effect of fuel economy standards on the U.S. automotive market: a hedonic demand analysis. *Transportation Research* 14A (5–6), 367–378.
- British Standards Institution, 2004. PD 7974-6, The application of fire safety engineering principles to fire safety design of buildings. Human factors. Life safety strategies. Occupant evacuation, behaviour and condition. London, BSI.
- Cardell, N., Dunbar, F., 1980. Measuring the societal impacts of automobile downsizing. *Transportation Research* 14A (5–6), 423–434.
- Cascetta, E., 1998. *Teoria e Metodi dell'Ingegneria dei Sistemi di Trasporto*. Turin, Utet.
- Cascetta, E., Nuzzolo, A., Biggiero, L., 1992. Analysis and modeling of commuters' departure time and route choice in urban networks. In: *Proceedings of the Second International Capri Seminar on Urban Traffic Networks*.
- Cirillo, E.N.M., Muntean, A., 2012. Can cooperation slow down emergency evacuations? *Comptes Rendus Mécanique* 340, 625–628.
- Candy, M.G., Chow, W.K., 2006. A brief review on the time line concept in evacuation. *International Journal on Architectural Science* 7, 1–13.
- Daamen, W., 2004. *Modelling Passenger Flows in Public Transport Facilities*, PhD Thesis, Technische Universiteit Delft, Netherlands.
- Deutsch, M., Gerard, H.B., 1955. A study of normative and informational social influences upon individual judgement. *Journal of Abnormal and Social Psychology* 5, 629–636.
- dell'Olio, L., Ibeas, A., Moura, J.L., 2009. Paying for parking: improving stated-preference surveys. *Proceedings of the ICE – Transport* 162 (1), 39–45.
- dell'Olio, L., Ibeas, A., Cecin, P., 2011. The quality of service desired by public transport users. *Transport Policy* 18, 217–227.
- Ehtamo, H., Heliövaara, S., Korhonen, T., Hostikka, S., 2010. Game theoretic best-response dynamics for evacuees' exit selection. *Advances in Complex Systems* 13, 113–134.
- Frantzich, H., 2001. Tid för utrymning vid brand. Time to evacuate in the event of fire. No. P21–365/01. Karlstad: Räddningsverket.
- Fridolf, K., 2010. *Fire Evacuation in Underground Transportation Systems: A Review of Accidents and Empirical Research*, PhD Thesis, Lund University, Sweden.
- Fridolf, K., Nilsson, D., Frantzich, H., 2011. Fire evacuation in underground transportation systems: a review of accidents and empirical research. *Fire Technology Journal*, 1–25.
- Galea R.G., Deere, S., Sharp, G., Filippidis, L., Hulse, L., 2010. Investigating the impact of culture on evacuation behaviour. In: *Proceedings of the 12th International Fire Science & Engineering Conference, Interflam 2010*, 5–7th July 2010, University of Nottingham, UK, Volume 1, pp. 879–892. ISBN 978 0 9541216-5-5, 2010.
- Gandit, M., Kouabenan, D.R., Caroly, S., 2008. Road-tunnel fires: risk perception and management strategies among users. *Safety Science* 47, 105–114.
- Greene, W., 2000. *Econometric Analysis*. McGraw Hill.
- Gwynne, S., Galea, E.R., Owen, M., Lawrence, P.J., Filippidis, L., 1999. A review of the methodologies used in evacuation modelling. *Building and Environment* 34 (6), 741–749.
- Hadjisophocleous, G.V., Bénichou, N., 2000. Development of performance-based codes, performance criteria and fire safety engineering methods. *International Journal on Engineering Performance-Based Fire Codes* 2 (4), 127–142.
- Helbing, D., Farkas, I.J., Molnar, P., Vicsek, T., 2002. Simulation of pedestrian crowds in normal and evacuation situations. In: Schreckenberg, M., Sharma, S.D. (Eds.), *Pedestrian and Evacuation Dynamics*. Springer, Berlin, pp. 21–58.
- Held, R.M., Durlach, N.I., 1992. Telepresence. In: *Presence: Teleoperators and Virtual Environments* 1, 109–112.
- Heliövaara, S., Kuusinen, J.M., Rinne, T., Korhonen, T., Ehtamo, H., 2012. Pedestrian behavior and exit selection in evacuation of a corridor – an experimental study. *Safety Science* 50 (2), 221–227.
- Heliövaara, S., 2007. *Computational Models for Human Behavior in Fire Evacuations*, MSc Thesis, Department of Engineering Physics and Mathematics, Helsinki University of Technology, Finland.
- Hensher, D.A., 2001. The valuation of commuter travel time savings for car drivers in New Zealand: evaluating alternative model specifications. *Transportation* 28 (2), 101–118.
- Hensher, D.A., Rose, J.M., Greene, W.H., 2005. *Applied Choice Analysis: A Primer*. Cambridge University Press, Cambridge.
- Hoogendoorn, S., Bovy, P., 2004. Pedestrian route-choice and activity scheduling theory and models. *Transportation Research Part B* 38, 169–190.
- Hoogendoorn, S.P., Bovy, P.H.L., Daamen, W., 2002. Microscopic pedestrian way finding and dynamics modelling. In: Schreckenberg, M., Sharma, S. (Eds.), *Pedestrian and Evacuation Dynamics*. Springer, Berlin, pp. 123–155.
- Johnson, C.W., 2005. Lessons from the evacuation of the world trade centre, September 11th 2001 for the development of computer-based simulations. *Cognition, Technology and Work* 7 (4), 214–240.
- Ibeas, A., dell'Olio, L., Montequin, R.B., 2011. Citizen involvement in promoting sustainable mobility. *Journal of Transport Geography* 19, 475–487.
- Kinateder, M., 2012. *Social Influence in Emergency Situations-Studies in Virtual Reality*. PhD Dissertation, University of Würzburg.
- Kocur, G., Adeler, T., Hyman, W., Aunet, B., 1982. Guide to forecasting travel demand with direct utility assessment, Report: UMTA-NH-11-0001-82-1Res. Rpt. 3.
- Korhonen, T., Hostikka, S., 2010. *Fire Dynamics Simulator with Evacuation: FDS+Evac*. Technical Reference and User's Guide, VTT Technical Research Centre of Finland, VTT Working Papers 119.
- Kuligowski, E.D., Peacock, R.D., Hoskins, B.L., 2010. A Review of Building Evacuation Models, second ed., NIST Technical Note 16806, National Institute of Standards and Technology, U.S. Department of Commerce, Washington, DC, USA.
- Kuligowski, E.D., 2011. Terror defeated: Occupant sense making, decision-making and protective action in the 2001 World Trade Center Disaster, PhD Thesis, University of Colorado, USA.
- Latané, B., Darley, J.M., 1968. *The Unresponsive Bystander: Why Does Not He Help?* Appleton-Century-Crofts, New York.
- Lo, S.M., Huang, H.C., Wang, P., Yuen, K.K., 2006. A game theory based exit selection model for evacuation. *Fire Safety Journal* 41 (5), 364–369.
- Lo, S.M., Fang, Z., Lin, P., Zhi, G.S., 2004. An evacuation model: the SGEM package. *Fire Safety Journal* 39 (3), 169–190.
- Lovreglio, R., Ronchi, E., Colonna, P., Berloco, N., 2013. A sensitivity analysis tool for road tunnel evacuation scenarios, *Tunnelling and Underground Space Technology Journal*, under review.
- Low, D., 2000. Following the crowd. *Nature* 407, 465–466.
- McLean, G.A., George, M.H., Funkhouser, G.E., Chittum, C.B., 1996. *Aircraft Evacuations Onto Escape Slides and Platforms I: Effects of Passenger Motivation*. Technical Report DOT/FAA/AM-96/18. FAA Civil Aeromedical Institute.
- McFadden, D., 1974. Conditional Logit Analysis of Qualitative Choice Behavior. In: Zarembka, P. (Ed.), *Frontiers in Econometrics*. Academic Press, New York.
- McFadden, D., Train, K., 2000. Mixed MNL models of discrete response. *Journal of Applied Econometrics* 15, 447–470.
- Meacham, B.J., 1998. The evolution of the Performance Based Code and Fire Safety Design Methods, Washington, NIST GCR-98-761.
- Muir, H., Cobbett, A., 1995. Cabin Crew Behaviour in Emergency Evacuations. Civil Aviation Authority/Federal Aviation Administration Paper DOT/FAA/CT-95/16.
- Nelson, H.E., Mowrer, F.W., 2002. Emergency Movement, In: Di Nenno, P.J., et al (Eds.), *SPPE Handbook of Fire Protection Engineering* 4th ed. Quincy: National Fire Protection Association, 2002, pp. 3–82-3-161.
- Nilsson, D., 2009. Exit Choice In Fire Emergencies – Influencing Choice Of Exit With Flashing Lights, PhD Thesis, Lund University, Sweden.
- Nilsson, D., Johansson, A., 2009. Social influence during the initial phase of a fire evacuation-analysis of evacuation experiments in a cinema theatre. *Fire Safety Journal* 44, 71–79.
- Norén, A., Winér, J., 2003. *Modelling Crown Evacuation from Road and Train Tunnels – Data and Design for Faster Evacuations*, Report 5127. Department of Fire Safety Engineering and Systems Safety, Lund University, Sweden.
- Pan, X., 2006. *Computational Modeling of Human and Social Behaviors for Emergency Egress Analysis*. Stanford University, California, Dissertation.
- Proulx, G., 1993. A stress model for people facing a fire. *Journal of Environmental Psychology* 13, 137–147.
- Purser, D.A., 2003. ASET and RSET: addressing some issues in relation to occupant behaviour and tenability. *Fire Safety Science* 7, 91–102. <http://dx.doi.org/10.3801/IAFSS.FSS.7-91>.
- Purser, D., Bensilum, M., 2001. Quantification of behavior for engineering design standards and escape time calculations. *Safety Science* 38, 157–182.
- Purser, D.A., Gwynne, S.M.V., 2007. Identifying Critical Evacuation Factors and the Application of Egress Models. In: *Proceedings of the 11th International* 79

- Interflam Conference, Franks, C.A., Grayson, S., (Eds.) pp. 203–214, London, England: Interscience Communications Ltd.
- Robin, T., 2011. New Challenges in Disaggregate Behavioral Modeling: Emotions, Investments and Mobility, PhD Thesis, École Polytechnique Fédérale de Lausanne, Switzerland.
- Ronchi, E., Colonna, P., Berloco, N., 2013. Reviewing Italian fire safety codes for the analysis of road tunnel evacuations: advantages and limitations of using evacuation models. *Safety Science* 52, 28–36.
- Ronchi, E., Nilsson, D., Gwynne, S.M.V., 2012a. Modelling the impact of emergency exit signs in tunnels. *Fire Technology* 48 (4), 861–988.
- Ronchi, E., 2012. Evacuation Modelling in Road tunnel Fires. Phd Thesis. Polytechnic University of Bari, Italy.
- Ronchi, E., Colonna, P., Capote, J., Alvear, D., Berloco, N., Cuesta, A., 2012b. The evaluation of different evacuation models for assessing road tunnel safety analysis. *Tunnelling and Underground Space Technology*.
- Santos, G., Aguirre, B.E., 2004. A Critical Review of Emergency Evacuation Simulation Models, NIST Workshop on Building Occupant Movement during Fire Emergencies June 9–10, National Institute of Standards and Technology. U.S. Department of Commerce, Washington, DC, USA.
- Sime, J.D., 1985. Movement towards the familiar – person and place affiliation in a fire entrapment setting. *Environment and Behaviour* 17 (6), 697–724.
- Tavares, R.M., 2009. An analysis of the fire safety codes in Brazil: is the performance-based approach the best practice? *Fire Safety Journal* 44, 749–755.
- Thurstone, L.L., 1927. A law of comparative judgment. *Psychological Review* 34, 273–286.
- Train, K., 1986. *Qualitative Choice Analysis: Theory, Econometrics and an Application to Automobile Demand*. Cambridge, MIT Press.
- Train, K., 1999. *Halton Sequence for Mixed Logit*. University of California, Berkeley.
- Train, K., 2009. *Discrete Choice Methods with Simulation*, second ed.. Cambridge University Press, Cambridge.
- Walker, J., Ben-Akiva, M., 2002. Generalized random utility model. *Mathematical Social Sciences* 43 (3), 303–343.
- Wilde, G.J.S., 2001. *Target Risk 2*. PDE Publications, Toronto, CA.
- Worm, E., 2006. *Human Behaviour Influencing Tunnel Safety*, Dutch Ministry of Transport, Public Works and Water Management, Tunnelling Department, Utrecht, NL. Head of the Centre for Tunnel Safety.